

Principles of Computer Security, Sixth Edition

Chapter 9 Solutions

Key Terms Quiz

1. network
2. datagram
3. routing
4. Media Access Control (MAC) address
5. subnet mask
6. topology
7. local area network (LAN)
8. protocol
9. three-way handshake
10. Network Address Translation (NAT)

Multiple-Choice Quiz

1. **A.** Layer 1 of the OSI model is the physical layer.
2. **B.** The User Datagram Protocol (UDP) is by definition a connectionless protocol.
3. **C.** The process that dynamically assigns an IP address to a network device is called the Dynamic Host Configuration Protocol (DHCP).
4. **B.** The three-way handshake is SYN, SYN/ACK, ACK.
5. **D.** The Internet Control Message Protocol (ICMP) is a control and information protocol used by network devices to determine such things as a remote network's availability and the length of time required to reach a remote network.
6. **B.** The protocol that translates names into IP addresses is known as Domain Name System (DNS).
7. **C.** Dividing the network address space into smaller networks is called subnetting.
8. **A.** The protocol that translates private (nonroutable) IP addresses into public (routable) IP addresses is called Network Address Translation (NAT).
9. **C.** The Transmission Control Protocol (TCP) carries sequence numbers, which allows for packets to be processed in the order they were sent.

10. C. The only entry that would represent a valid MAC address is 00:07:e9:7c:c8:aa. All the others are either too short or too long.

Essay Quiz

1. Answers to this question will vary but should include a basic discussion of the advantages and disadvantages of the TCP and UDP protocols. Some students may present a greater knowledge than what is presented in the text, but all answers should include the following:

TCP Protocol

- **Advantages** Connection oriented; guaranteed delivery; packets processed in sequence
- **Disadvantages** Requires overhead; higher cost in terms of bandwidth and processing

UDP Protocol

- **Advantages** Connectionless; lightweight and fast; good for streaming media
- **Disadvantages** No guarantees; no error checking; packets may be processed out of sequence

2. Answers to this question will vary but should include a basic discussion of DHCP with the following points:

- Hosts are dynamically assigned an IP address when they connect to the network.
- IP addresses are assigned with a limited lifespan.
- The system must request another IP address when its DHCP lease expires.
- IP addresses are assigned from an available pool, and no address is given if no more addresses are available.

For the opinion side of the question, answers must be judged on how well students defend their position. Most students should see that DHCP is really only appropriate for client systems. Servers are much more likely to have static IP addresses assigned to each server.

3. Answers should include a discussion of each of the three main topologies discussed in this chapter:

- **Star topology** All network devices are connected to a central point. This creates a star-like pattern for which the topology is named, as shown in Figure 9.2 of the textbook.

- **Bus topology** All network components are connected to the same cable, often called “the bus” or “the backbone,” as shown in Figure 9.3 of the textbook.
 - **Ring topology** Network components are connected to each other in a closed loop, with each device directly connected to two other devices, as shown in Figure 9.4 of the textbook.
4. Answers should discuss the three-way handshake used to initiate TCP connections between two systems. The initiating system sends a SYN to the system it needs to open a connection with. The responding system sends back a SYN/ACK to the initiating system. The initiating system sends back an ACK to complete the three-way handshake and initialize the TCP connection. Some students may also provide a diagram, such as the one shown in Figure 9.12 of the textbook.
5. Answers to this question will vary widely but all should include the following key points:
- IP addresses are 32-bit numbers that are broken up into a network portion and a host portion.
 - The network portion of the IP address is also known as the network ID.
 - A subnet mask is used to tell you how much of the 32-bit address space is being used to represent the network portion of the address and how much is being used to represent the host portion of the address.

The examples will likely follow the sample from the chapter:

Subnet mask: 255.255.255.0

IP address: 10.10.10.101

First, you must convert the address and subnet mask to their binary representations:

Subnet mask: 11111111.11111111.11111111.00000000

IP address: 00001010.00001010.00001010.01100101

Then, you perform a bitwise AND operation to get the subnet address. The bitwise AND operation examines each set of matching bits from the binary representation of the subnet mask and the IP address. For each set where both the mask and address bits are 1, the result

is a 1. Otherwise, if either bit is a 0, the result is a 0. So, for this example we get the following:

Network ID: 00001010.00001010.00001010.00000000